

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Advanced Math

05

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**B****B. Exactly three**

Choice B is correct. The given equation can be rewritten as

$52(x + 4)(x^2 - 4x + 16)(x^2 - 9)(x^2 + 9) = 0$, which is equivalent to

$52(x + 4)(x^2 - 4x + 16)(x - 3)(x + 3)(x^2 + 9) = 0$. Applying the zero product property in this equation yields $52 = 0$, $x + 4 = 0$, $x^2 - 4x + 16 = 0$, $x - 3 = 0$, $x + 3 = 0$, and $x^2 + 9 = 0$. The equation $52 = 0$ has no solution. Subtracting 4 from both sides of the equation $x + 4 = 0$ yields $x = -4$. Adding 3 to both sides of the equation $x - 3 = 0$ yields $x = 3$. Subtracting 3 from both sides of the equation $x + 3 = 0$ yields $x = -3$. The discriminant of a quadratic equation of the form $ax^2 + bx + c = 0$ is $b^2 - 4ac$. If the discriminant is positive, the equation has two distinct real solutions. If the discriminant is zero, the equation has one distinct real solution. If the discriminant is negative, the equation has no real solutions. In the equation $x^2 - 4x + 16 = 0$, $a = 1$, $b = -4$, and $c = 16$. Substituting these values in the equation for the discriminant, $b^2 - 4ac$, yields $(-4)^2 - 4(1)(16)$, or -48 . Since the discriminant is negative, this equation has no real solutions. In the equation $x^2 + 9 = 0$, $a = 1$, $b = 0$, and $c = 9$. Substituting these values in the equation for the discriminant, $b^2 - 4ac$, yields $0^2 - 4(1)(9)$, or -36 . Since the discriminant is negative, this equation has no real solutions. Therefore, the distinct real solutions to the given equation are $x = -4$, $x = 3$, and $x = -3$. Thus, the given equation has exactly three distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**D**

D. $7x^4 + 9x^3$

Choice D is correct. Adding the two expressions yields $3x^4 + 2x^3 + 4x^4 + 7x^3$. Because the pair of terms $3x^4$ and $4x^4$ and the pair of terms $2x^3$ and $7x^3$ each contain the same variable raised to the same power, they are like terms and can be combined as $7x^4$ and $9x^3$, respectively. The sum of the given expressions therefore simplifies to $7x^4 + 9x^3$.

Choice A is incorrect and may result from adding the coefficients and the exponents in the given expressions. Choice B is incorrect and may result from adding the exponents as well as the coefficients of the like terms in the given expressions. Choice C is incorrect and may result from multiplying, rather than adding, the coefficients of the like terms in the given expressions.

Q3**D**

D. $gx = \frac{1}{25}^x$

Choice D is correct. It's given that function g is exponential. Therefore, an equation defining g can be written in the form $gx = ab^x$, where a and b are constants. The table shows that when $x = 0$, $gx = 1$. Substituting 0 for x and 1 for gx in the equation $gx = ab^x$ yields $1 = ab^0$, which is equivalent to $1 = a$. Substituting 1 for a in the equation $gx = ab^x$ yields $gx = b^x$. The table also shows that when $x = 1$, $gx = \frac{1}{25}$. Substituting 1 for x and $\frac{1}{25}$ for gx in the equation $gx = b^x$ yields $\frac{1}{25} = b^1$, which is equivalent to $\frac{1}{25} = b$. Substituting $\frac{1}{25}$ for b in the equation $gx = b^x$ yields $gx = \frac{1}{25}^x$.

Choice A is incorrect. For this function, $g1$ is equal to -25 , not $\frac{1}{25}$.

Choice B is incorrect. For this function, $g1$ is equal to $-\frac{1}{25}$, not $\frac{1}{25}$.

Choice C is incorrect. For this function, $g1$ is equal to 25 , not $\frac{1}{25}$.

Q4**B**

B. $a = \frac{x}{8b+9}$

Choice B is correct. To express a in terms of b and x , the given equation can be rewritten such that a is isolated on one side of the equation. Since it's given that b is a positive number, $b + 9$ is not equal to zero. Therefore, dividing both sides of the given equation by $8b + 9$ yields the equivalent equation $\frac{x}{8b+9} = a$, or $a = \frac{x}{8b+9}$.

Choice A is incorrect. This equation is equivalent to $x = 8a + b + 9$.

Choice C is incorrect. This equation is equivalent to $x = \frac{8b+9}{a}$.

Choice D is incorrect. This equation is equivalent to $x = \frac{a}{8b+9}$.

Q5**A**

A. 7

Choice A is correct. Since the given expressions are equivalent and the numerator of the second expression is $\frac{1}{6}$ of the numerator of the first expression, the denominator of the second expression must also be $\frac{1}{6}$ of the denominator of the first expression. By the distributive property, $\frac{1}{6}6x + 42$ is equivalent to $\frac{1}{6}6x + \frac{1}{6}42$, or $x + 7$. Therefore, the value of b is 7.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**D**

D. $p = 15,000(1.04)^x$

Choice D is correct. It's given that a model predicts the population of Bergen in 2005 was 15,000. The model also predicts that each year for the next 5 years, the population increased by 4% of the previous year's population. The predicted population in one of these years can be found by multiplying the predicted population from the previous year by 1.04. Since the predicted population in 2005 was 15,000, the predicted population 1 year later is $15,000(1.04)$. The predicted population 2 years later is this value times 1.04, which is $15,000(1.04)^2$, or $15,000(1.04)^2$. The predicted population 3 years later is this value times 1.04, or $15,000(1.04)^3$. More generally, the predicted population, p , x years after 2005 is represented by the equation $p = 15,000(1.04)^x$.

Choice A is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 0.96 rather than 15,000.

Choice B is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 1.04 rather than 15,000.

Choice C is incorrect. This equation indicates the predicted population is decreasing, rather than increasing, by 4% each year.

Q7**1**

The correct answer is 1. The point of intersection of the graphs of the given equations is the solution to the system of the two equations. Since $y = (x - 1)^2$ and $y = 2x - 3$, it follows that $(x - 1)^2 = 2x - 3$, or $(x - 1)(x - 1) = 2x - 3$. Applying the distributive property to the left-hand side of this equation yields $x^2 - 2x + 1 = 2x - 3$. Subtracting $2x$ from and adding 3 to both sides of this equation yields $x^2 - 4x + 4 = 0$. Factoring the left-hand side of this equation yields $(x - 2)(x - 2) = 0$. By the zero product property, if $(x - 2)(x - 2) = 0$, it follows that $x - 2 = 0$. Adding 2 to both sides of $x - 2 = 0$ yields $x = 2$. Substituting 2 for x in either of the given equations yields $y = 1$. For example, substituting 2 for x in the second given equation yields $y = 2(2) - 3$, or $y = 1$. Therefore, the point of intersection of the graphs of the given equations is $(2, 1)$. The y-coordinate of this point is 1.

Q8**B**

B. $-2x^3 - 7x^2 + x + 3$

Choice B is correct. Using the distributive property, the given expression can be rewritten as $4x^3 - 5x^2 + 3 - 6x^3 - 2x^2 + x$. Combining like terms, this expression can be rewritten as $(4 - 6)x^3 + (-5 - 2)x^2 + x + 3$, which is equivalent to $-2x^3 - 7x^2 + x + 3$.

Choices A, C, and D are incorrect and may result from an error when applying the distributive property or an error when combining like terms.

Q9**B**

B. 790

Choice B is correct. It's given that t minutes after an initial observation, the number of bacteria in a population is $40,000 \cdot 2^{\frac{t}{790}}$. This expression consists of the initial number of bacteria, 40,000, multiplied by the expression $2^{\frac{t}{790}}$. The time, in minutes, it takes for the number of bacteria to double is the increase in the value of t that causes the expression $2^{\frac{t}{790}}$ to double. Since the base is 2, the expression $2^{\frac{t}{790}}$ will double when the exponent increases by 1. Since the exponent of this expression is $\frac{t}{790}$, the exponent will increase by 1 when t increases by 790. Therefore, the time, in minutes, it takes for the number of bacteria in the population to double is 790.

Choice A is incorrect. This is the base of the exponent, not the time it takes for the number of bacteria in the population to double.

Choice C is incorrect. This is the number of minutes it takes for the population to double twice.

Choice D is incorrect. This is the number of bacteria that are initially observed, not the time it takes for the number of bacteria in the population to double.

Q10

A

A. Exactly two

Choice A is correct. The number of solutions of a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $x^2 - 12x + 27 = 0$, $a = 1$, $b = -12$, and $c = 27$. Substituting these values for a , b , and c in $b^2 - 4ac$ yields $-12^2 - 4(1)(27)$, or $-36 - 108$, or -144 . Since the value of its discriminant is negative, the given equation has zero real solutions.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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